



School of Computer Science Engineering and Information Systems

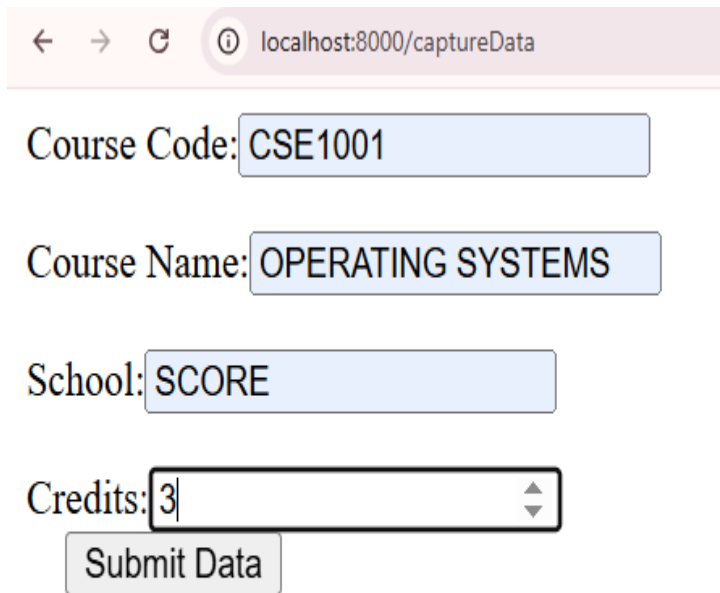
Full Stack Web Development –Lab NodeJS

CourseId : PAMCA506

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DATE GIVEN: 18/2/2026

1. Assume that the server-side Node.js script maintains the details of various employees in an array. Employee details include EMPLOYEE NAME, DEPARTMENT, and SALARY. Design a form to select various employees and facilitate total salary calculation at the server end upon submitting the form.
2. Design a single Node.js web application with different URLs to manage the session process as follows:
 - a. A form to collect the session key and value.
 - b. A button to either create a new session variable if it does not exist or update its value if it already exists. Display the corresponding details.
 - c. A button to list all active session variables, displaying each session key and its value when clicked.
3. Develop a Node JS script to read a file name from a query string and execute a read operation for a time interval of every 200 milliseconds. Display the file content along with number of lines and words present in it.
4. Using Node JS develop a user interface to capture the details of course like Course Code, Course Name, School offering the Course and Credits. Implement the below using cookie parser API
 - On submission of form, the server should send back the course details and the metadata information about server (refer Figure 3) as Cookie and redirecting to the route displaying the details of course as shown in Figure 2.
 - On Clicking “List All Cookies” hyperlink display all cookies as shown in Figure 3

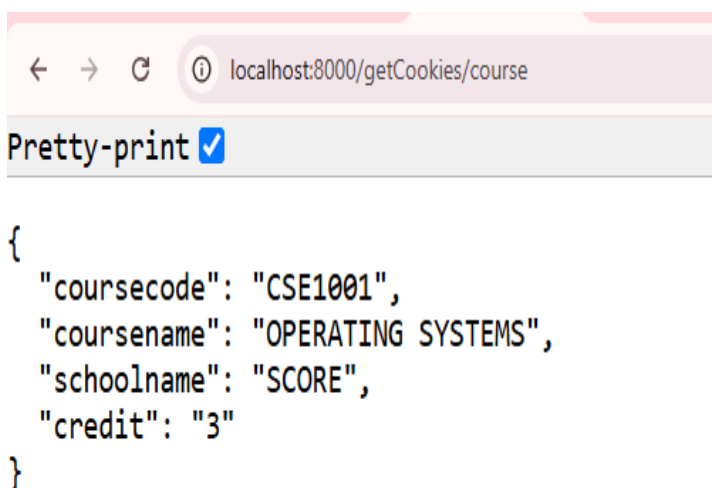


A screenshot of a web browser window. The address bar shows 'localhost:8000/captureData'. The form contains four input fields: 'Course Code' with the value 'CSE1001', 'Course Name' with the value 'OPERATING SYSTEMS', 'School' with the value 'SCORE', and 'Credits' with the value '3'. Below the 'Credits' field is a 'Submit Data' button.

[LIST ALL COOKIES](#)

Figure 1. Landing Page

After clicking ‘Submit Data’ below screen should be displayed



A screenshot of a web browser window. The address bar shows 'localhost:8000/getCookies/course'. Below the address bar is a 'Pretty-print' checkbox which is checked. The main content area displays a JSON object representing the course details.

```
{
  "coursecode": "CSE1001",
  "coursename": "OPERATING SYSTEMS",
  "schoolname": "SCORE",
  "credit": "3"
}
```

Figure 2. Course Details as Cookie Response

On clicking “List All Cookies” hyperlink below content should be displayed

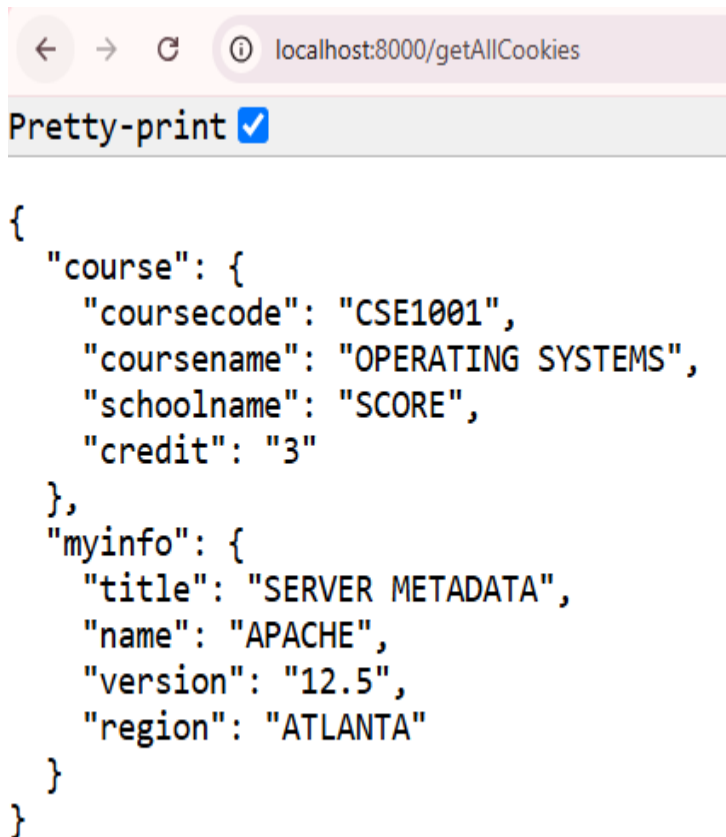


Figure 3. Display all Cookies.

5. Assume that the server side Node JS script maintains the details of various products in an array. Product details include PRODUCT NAME, COLOR and PRICE. Design the form to select the various products and facilitate total bill generation at the server end on submitting the form.

CLIENT FORM

Product Selection

Select Product:

☐ Dinner set ☒ Fashion set ☒ Table set ☐ Fabric set

GENERATE BILL

After clicking 'Generate Bill' button, display the total bill as shown below:



Total Price = 5788 INR

[CLICK HERE TO ORDER AGAIN](#)

6. Develop a Node JS application to authenticate the credentials data supplied through HTML form using the JSON array maintained in the script. Ensure JSON array consists of keys named UserID, Password, Username and Failure_Attempt.

During each unsuccessful attempt increment Failure_Attempt by 1 and display the login page.

In case Failure_Attempt reaches 3, display warning message alone. (Use NodeJS session)

7. Write a Node.js application that allows users to select a file through a user interface and click the 'UPLOAD' button. Validate that the selected file type is 'pdf' before uploading it to the server. Finally, display a message indicating whether the file is a valid PDF and if it was uploaded successfully.
